

ZHERDEVKA, Russian FederationWMO: **340470**Lat: **51.833N**Lon: **41.483E**Elev: **147**StdP: **99.57**Time Zone: **3.00 (E03)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-25.7	-22.3	-29.2	0.3	-25.5	-25.5	0.4	-22.0	11.7	-3.2	10.0	-6.0	2.5	300	0.544

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.7	31.7	19.5	29.8	18.9	28.1	18.2	21.2	28.6	20.3	27.0	19.5	25.7	3.3	120

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.9	14.0	23.9	18.1	13.2	23.0	17.3	12.6	22.2	62.1	28.8	59.1	27.3	56.4	25.7	24.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.3	8.0	7.0	-27.1	34.6	2.9	2.2	-29.2	36.2	-30.9	37.5	-32.5	38.7	-34.6	40.2		
(5)			-27.8	22.6	3.5	1.1	-30.3	23.4	-32.3	24.0	-34.2	24.6	-36.8	25.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	6.6	-8.2	-7.7	-2.2	7.9	15.3	18.8	20.5	19.5	13.9	6.8	-0.6	-5.7		
	DBStd	11.79	6.81	6.95	5.08	4.95	5.03	4.07	3.08	3.67	4.40	4.93	5.81	6.52		
	HDD10.0	2492	563	497	379	97	13	1	0	0	14	124	319	487		
	HDD18.3	4524	821	730	637	313	121	45	13	30	142	359	568	745		
	CDD10.0	1250	0	0	0	34	177	264	324	294	132	24	1	0		
	CDD18.3	239	0	0	0	0	26	59	79	66	10	0	0	0		
	CDH23.3	2368	0	0	0	7	272	559	736	677	117	1	0	0		
	CDH26.7	714	0	0	0	0	65	167	238	226	18	0	0	0		
(14) Wind	WSAvg	3.3	3.7	3.9	3.9	3.7	3.1	2.8	2.6	2.7	2.8	3.2	3.6	3.9		
(15) Precipitation	PrecAvg	502	36	30	27	34	45	67	56	37	58	44	39	38		
	PrecMax	661	71	56	55	132	101	151	157	127	141	90	88	73		
	PrecMin	342	11	8	3	5	11	21	3	3	3	13	2	13		
	PrecStd	89	14	14	16	29	25	36	32	28	40	23	28	16		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	2.6	3.8	12.2	24.5	30.5	32.4	34.5	33.9	28.6	21.7	12.7	7.7		
		MCWB	1.5	2.9	7.4	12.8	18.5	19.0	20.6	20.2	16.2	13.6	11.1	6.9		
	2%	DB	1.6	2.2	8.5	21.6	28.6	30.3	31.8	31.4	26.6	18.7	10.3	3.9		
		MCWB	0.8	1.3	5.5	12.5	17.1	19.4	20.3	19.3	15.8	12.4	9.1	3.1		
	5%	DB	0.9	1.5	6.0	19.1	26.6	28.6	29.2	29.1	24.3	16.1	8.4	2.2		
		MCWB	0.4	0.7	3.7	11.4	16.7	19.0	19.2	18.2	15.3	11.8	7.3	1.5		
	10%	DB	0.3	0.8	4.0	16.7	24.2	26.7	27.4	27.1	21.8	13.8	6.4	1.2		
		MCWB	-0.2	0.1	2.3	10.0	15.8	18.5	18.6	17.9	14.4	10.6	5.4	0.6		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	1.9	3.0	8.3	15.0	20.0	22.1	22.6	22.2	18.9	15.7	11.6	6.8		
		MCDB	2.2	3.7	11.2	20.1	27.2	28.5	30.3	29.4	25.3	19.5	12.7	7.7		
	2%	WB	1.0	1.6	6.0	13.3	18.6	21.0	21.1	20.8	17.3	14.0	9.3	3.3		
		MCDB	1.3	1.9	8.3	18.9	26.2	27.7	28.6	28.9	22.3	17.1	10.0	3.9		
	5%	WB	0.5	0.8	4.0	12.0	17.6	20.0	20.3	19.7	16.2	12.3	7.4	1.6		
		MCDB	0.9	1.3	5.8	18.2	24.4	25.8	27.0	26.3	21.8	15.3	8.2	2.0		
	10%	WB	-0.2	0.2	2.5	10.7	16.4	19.0	19.5	18.8	15.1	10.8	5.6	0.7		
		MCDB	0.3	0.8	3.8	16.3	22.8	25.2	25.4	25.0	20.9	13.4	6.4	1.2		
(35) Mean Daily Temperature Range	5% DB	MDBR	5.6	6.5	6.8	10.0	11.0	10.3	10.7	11.6	10.4	7.5	5.1	5.1		
		MCDBR	3.5	4.3	8.5	13.9	13.7	13.1	13.6	14.5	14.2	11.8	6.8	4.0		
	5% WB	MCWBR	3.5	4.1	6.0	7.1	5.5	5.2	5.0	5.1	6.1	6.9	6.0	3.8		
		MCWBR	3.6	4.3	8.1	12.4	12.1	11.0	11.3	12.1	11.6	10.1	6.2	4.1		
(40) Clear-Sky Solar Irradiance	taub		0.304	0.341	0.366	0.434	0.398	0.390	0.417	0.438	0.386	0.362	0.338	0.308		
	taud		2.108	2.007	2.070	2.105	2.314	2.355	2.278	2.209	2.340	2.348	2.260	2.148		
	Ebn at Noon		665	739	810	798	856	865	831	788	791	732	631	591		
	Edh at Noon		76	113	129	142	121	117	125	127	100	81	67	63		
(44) All-Sky Solar Radiation	RadAvg		0.92	1.93	3.41	4.33	5.80	6.02	5.94	5.07	3.40	1.76	0.78	0.52		
	RadStd		0.21	0.31	0.52	0.35	0.50	0.56	0.47	0.43	0.41	0.27	0.17	0.14		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (7 neighbors)		+0.70	N/A	N/A	N/A	N/A	N/A	N/A	-137	+122	N/A

Nomenclature: See separate page